A WEAKLY ETALE UNIVERSAL HOMEOMORPHISM OF SCHEMES IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19189-19193 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19195-19208, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Closing card of the weakly etale trilogy; together with L16089-L16090 it forms the chapter's self-contained weakly etale toolkit.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms-chapter facts that a universal homeomorphism is submersive with universally injective underlying map and that a flat closed immersion is an open-and-closed immersion/isomorphism on components; Schemes, lemma monomorphism (diagonal iso iff monomorphism); Descent, lemma flat-surjective-quasi-compact-monomorphism-isomorphism.

Source reference. Original lemma 06NC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19208–19208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04PW: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19208–19208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04DE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19208–19208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01S4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19208–19208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01L3: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19208–19208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. If f is weakly étale and a universal homeomorphism, it is an isomorphism.*

Proof. Since f is a universal homeomorphism, the diagonal $\Delta : X \to X \times_Y X$ is a surjective closed immersion by Morphisms, Lemmas 04DE and 01S4. Since Δ is also flat, we see that Δ must be an isomorphism by Morphisms, Lemma 04PW. In other words, f is a monomorphism (Schemes, Lemma 01L3). Since f is a universal homeomorphism it is certainly quasi-compact. Hence by Descent, Lemma 06NC we find that f is an isomorphism. □